

CAPRM-Flood: A Reproducible C++/Python Framework for Property-Level Flood Exposure Indexing

Student: Sterling Alexander
Advisor: Professor Minseok Kwon

Abstract. CAPRM-Flood is a reproducible C++/Python geospatial feature-extraction framework for large-batch property-to-hazard spatial joins, demonstrated through a property-level flood exposure indexing workload for Monroe County / Rochester, NY. Given a table of property identifiers and coordinates, the system will derive public-data features such as FEMA flood-zone membership, nearest-water distance, elevation or terrain indicators, precipitation-frequency indicators, and regional context from public hazard, hydrography, elevation, precipitation, parcel, and boundary datasets. These features will be normalized into interpretable component scores and combined using explicit configurable weights into a transparent 0–100 Relative Flood Exposure Index. The project does not attempt to predict flood losses, insurance premiums, property prices, or actuarial risk. Its primary computer science contribution is the design, implementation, and evaluation of a reproducible geospatial data-processing system, including a performance-critical C++ spatial-processing core, Python orchestration and scoring modules, correctness validation against standard Python geospatial baselines, runtime and memory benchmarking across increasing dataset sizes, and documented assumptions, limitations, and reproducibility practices.